a)

x	y	Forward slopes
0	40	// / /
1	44.39	$\frac{44.39-40}{1} = 4.39$
2	47.169	$47.169-44.39 = 2.779$
3	48.463	$48.463-47.169 = 1.294$
4	49.069	0.606
5	50.077	1.008
6	52.273	2.146
7	55.77	3.497
8	60.152	4.382

↳ At boundaries, you run out of reference points; index error

b)

$x \in$	$y = Ax^2+Bx+C$	
[0, 1.5]	$-0.8055x^2+5.1955x+40$	Using $x=0, x=1, x=2$
[1.5, 2.5]	$-0.7425x^2+5.0065x+40.1260$	Using $x=1, 2, 3$
[2.5, 3.5]	$-0.3440x^2+3.0140x+42.5170$	$x=2, 3, 4$
⋮	$0.2010x^2-0.8010x+49.0570$	
	$0.6505x^2-4.3380x+56.9170$	
	$0.4425x^2-2.2555x+49.8760$	
[7.5, 8]	$0.4425x^2-2.2555x+49.8760$	

c) $\left.\frac{dy}{dx}\right|_{x=x_{center}}$: [0, 1.5] [1.5, 2.5]

3.5845 | 2.0365 | 0.95 | 0.8070 | 1.6020 | 2.8465 |

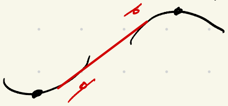
3.9395

↳ Responds faster, But Close Enough for 8 points

↳ Smoother at nodes.

d) & centered the quadratic around an x-value.
↳ Sliding window depends on "length" scale / weight / significance of each data point
↳ Fixed Assumes Equal Contribution to response.
↳ if you choose outlier, center quadratic around it, you create a "lie".

e) Just like tangent lines to curves in SOLIDWORKS sketches



→ fillet both lines w/ r=0
↳ if Overlapping, truncate and enforce continuity.

f) Just like anything, noise creates confidence intervals, which is no good.
↳ Polynomials will have error that scales quadratically
↳ Their derivatives have more error than regular slopes
↳ Numerical estimates can do better, ANOVA CIs
↳ Numerical Diff. is more sensitive?
↳ Scaling order down numerically

g) Static Noise Reduction
↳ Delete all data that isn't same slope as expected / first slope
RK4
↳ better algorithm for derivatives (given data probably not fractals)
↳ Assumes smoother function, less sensitive to spikes

Smoother = less spikey = miss peaks
Preservation = less smooth = Poor interpolations

---

2. a) → Kernel assigns weight to data points, cov. builds map of effect each var has on one another.
→ Predictions are made w/ confidence % at each new point of mean function.

→ Uncertainty is quantified using the posterior

↳ Comes from kernel / Cov. matrix.